S1 Ablation Study for Activation Sparsity in Pythia-14M

Architecture, threshold gates, learned thresholds, pressure, and seed sensitivity

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Answer in brief.

- **Topology sets reach.** At the common AdamW setting, R_{model} rises from effectively 0% for stock A0 to 5.80% for the Three-ReLU A3 path and 13.64% for A6-POST. Adding V to matched Q/K parents contributes about 3.77 percentage points (pp) for less than 0.002 loss.
- **Thresholds expose a controllable tradeoff.** Increasing fixed κ raises R_{model} in every complete A5/A6 ladder, with a small but nonzero loss cost. Signed $G^{\pm}$ gates preserve lower loss than one-sided G^{+} gates, but give up 1.64–6.61 R_{model} pp. Learned absolute thresholds add opportunity in all 15 matched fixed-threshold contrasts; RMS-relative learning instead shifts toward lower loss and lower opportunity.
- **Pressure and orthogonalization are distinct controls.** OL1/OR have lower loss than their exact naive partners in all 20 seed-0 comparisons, but generally produce a smaller sparsity response. The 0.5 orthogonal-step cap binds at every logged A6-POST update, so these are results for a budget-constrained procedure.
- **The evidence is screening evidence.** One coupled seed change preserves all ten logical-compute effect signs and nine of ten loss-effect signs. S1 supports a matched scale-up panel; it does not establish variance, model-scale generality, sparse-kernel speedup, or a global winner.

1 Scope and Scientific Questions

Table 1: S1 experimental blocks. Every cell is a 2,048-step randomly initialized Pythia-14M pretraining run; block-local departures from the common contract are stated in the setup column.

Block	Scientific question	Controlled setup	Primary comparison
B0	How do ordinary-ReLU topology and model learning rate change quality and logical opportunity?	AdamW, no pressure; ten topologies; three LRs for five anchors; 20 cells.	Direct topology parent and within-architecture LR.
B1	How do fixed threshold, gate family, PRE/POST placement, and gated topology change quality and logical opportunity?	AdamW at 3×10^{-5}; A1-H through A6 variants; 36 positive-threshold cells.	Absolute endpoints by architecture, gate, and κ, including registered $\kappa = 0$ controls.
B2	Does learned κ move beyond fixed $\kappa = 0.10$, and how do score scale and threshold parameterization matter?	AdamW without pressure; 26 cells spanning architecture, gate, absolute/RMS-relative score, and three parameterizations.	Absolute endpoints for fixed versus learned and absolute versus RMS-relative thresholds.
B3	How do pressure family, orthogonalization, weight, and Ricker geometry change matched quality and opportunity?	A3 and A6-POST; all active gates; orthogonal variants use fixed step budget 0.5.	Pressure–AdamW and orthogonal–naive; 40 cells.
B4	Which selected endpoints and effect directions survive one coupled seed change?	Ten sentinels; initialization/data-order seeds 0/0 versus 1/1.	Endpoint shifts and within-seed sign agreement.

S1 contains 132 completed scientific cells and 36 pooled diagnostics. Two declared post-PV context-gate cells remained dependency-gated. The study is a mostly one-seed feasibility screen on one frozen selection set: comparisons are local and prespecified, no equivalence margin or multiplicity correction was defined, and no global S1 rank is treated as confirmatory.

2 Common Experimental Setup and Metrics

Table 2: Common S1 experimental contract. A cache-pass equivalent is a token ratio, not unique-token coverage or an epoch, because blocks are sampled with replacement.

Item	Frozen setting
Model	Randomly initialized `EleutherAI/pythia-14m-deduped`; released weights are not loaded. Six parallel-residual blocks, $d = 128$, MLP width 512, four heads, and sequence length 2,048.
Training data	MiniPile frozen train cache: 1,491,711,416 tokens. Random contiguous blocks with replacement.
Budget and batching	2,048 updates; 65,536 tokens/update; 134,217,728 tokens per cell = 0.089976 cache-pass equivalents (8.998%). Microbatch four sequences, gradient accumulation eight, effective batch 32 sequences.
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 0.01, no gradient clipping; 100-step linear warmup then constant LR. The common LR is 3×10^{-5}; only B0 has 10^{-5} and 10^{-4} flanks.
Validation	Frozen 250-document selection half of MiniPile validation: 311,739 cached tokens. Evaluation retains 152 complete sequences, 311,296 input positions, 38 batches, and discards 443 trailing tokens. Loss averages 311,144 shifted next-token targets.
Seeds and precision	Primary initialization/data-order seeds 0/0; B4 changes both to 1/1. FP32 parameters with bfloat16 autocast.
Execution and scale	Dense eager PyTorch on one NVIDIA GeForce RTX 5070 Ti Laptop GPU (12,227 MiB). The 132 cells sum to 17.72B scheduled tokens and 50.76 scientific-run GPU-hours. No sparse kernel is used.

Pythia uses parallel attention and MLP residual branches. For residual input H_ℓ, the relevant dataflow is

$$[\widehat{Q},\widehat{K},\widehat{V}] = A_\ell W_{\mathrm{QKV}} + b_{\mathrm{QKV}}, \quad P = \mathrm{softmax}(QK^\top/\sqrt{d_h} + \mathcal{M}), \quad C = PV,$$
$$H_{\ell+1} = H_\ell + (CW_o + b_o) + \big(\phi(M_\ell W_1 + b_1)W_2 + b_2\big),$$

where active gates may transform the post-LayerNorm branch inputs A_ℓ, M_ℓ, MLP hidden h, or post-projection Q/K/V. PRE gates act on Q/K before RoPE; POST gates act after RoPE. The final model LayerNorm is never gated.

The report uses two endpoint outcomes and one exact-zero counting convention:

- L: final step-2,048 selection next-token cross-entropy.
- An *exact zero*: direct floating-point equality $x = 0$, with no tolerance. Counts are integer-pooled over the 152 validation sequences, all applicable coordinates, and all six layers.
- $R_m \equiv R_{\mathrm{model}}$: the model-wide share of logical scalar products with at least one exact-zero activation operand in QKV, causal-valid QK, causal-valid PV, W_o, W_1, or W_2. Future causal positions are excluded and the LM head is treated as dense.

Writing the exact-zero product count as Z,

$$R_{\mathrm{model}} = \frac{Z}{857{,}085{,}050{,}880 + 2{,}004{,}407{,}549{,}952},$$

where the first denominator term covers all six transformer blocks and the second covers the LM head. R_m is a logical opportunity, not measured FLOPs, latency, energy, or sparse-kernel speedup.

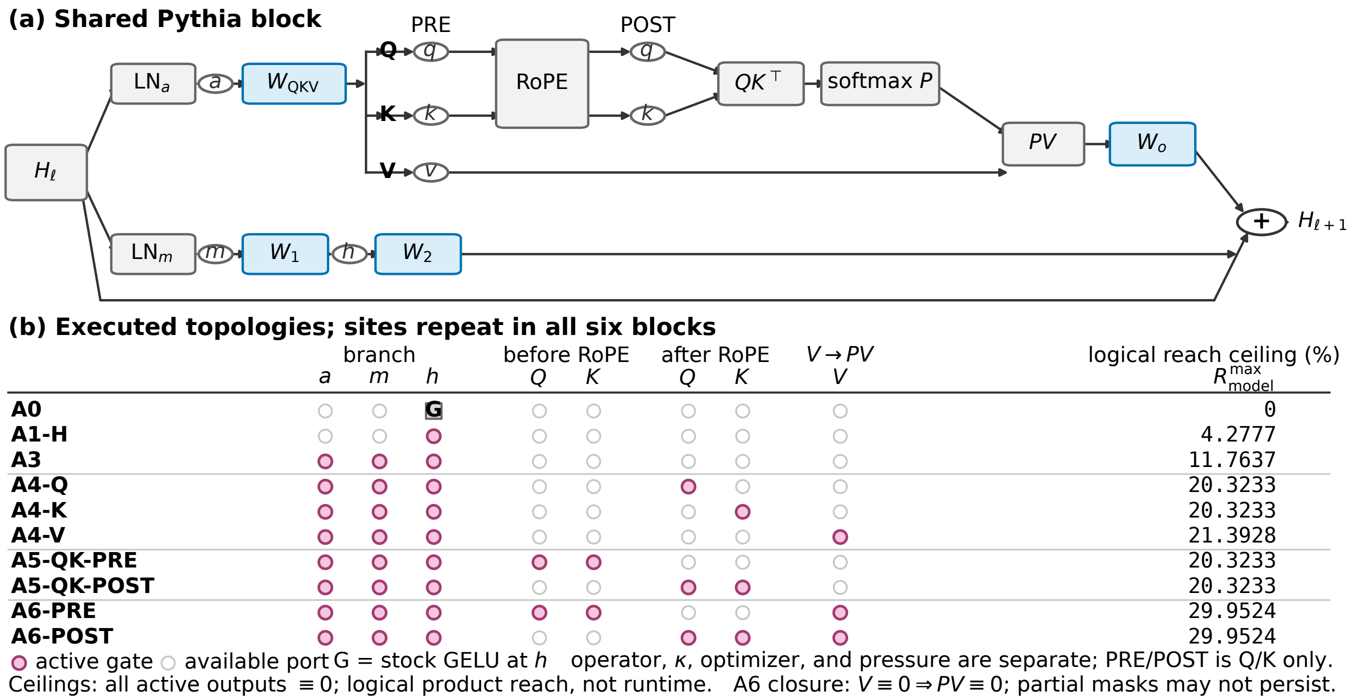

Figure 1: Pythia-14M topology atlas. Each row is one parallel-residual block; filled gate markers show the exact-zero paths enabled by that architecture. The right column gives the limiting model-wide reachable share under complete zeroing, not the observed partial-mask values. PRE/POST refers to Q/K placement around RoPE; V is always pre-PV.

3 Experimental Blocks

3.1 B0: Learning Rate and Architecture

Question and setup. B0 asks how ordinary exact-zero paths and the three tested model LRs change L and R_m. A0 retains stock GELU; every gated topology uses ordinary one-sided $G_0^+(x) = \max(x, 0)$. All 20 cells use AdamW without activation pressure.

Table 3: Complete B0 endpoints with learning rates in parallel column groups. Each group reports validation loss and R_m (%). All 20 cells use seed-0 AdamW for 2,048 steps without activation pressure. A0 uses stock GELU; gated architectures use ordinary G_0^+. A dash means that the architecture was not executed at that learning rate. Config IDs remain in the complete appendix.

Architecture	10^{-5}		3×10^{-5}		10^{-4}	
	L	R_m (%)	L	R_m (%)	L	R_m (%)
A0	8.35104	0.00	7.04913	0.00	5.93887	0.00
A1-H	8.38652	1.97	6.98875	2.08	5.87474	2.40
A3	8.40015	5.69	7.01310	5.80	5.91768	6.16
A4-Q	–		7.02126	7.71	–	
A4-K	–		7.02441	8.28	–	
A4-V	–		7.01615	9.50	–	
A5-QK-PRE	–		7.01586	8.45	–	
A5-QK-POST	–		7.03064	9.87	–	
A6-PRE	8.38739	13.80	7.01645	12.22	5.93539	12.37
A6-POST	8.38743	14.83	7.03248	13.64	6.06320	13.64

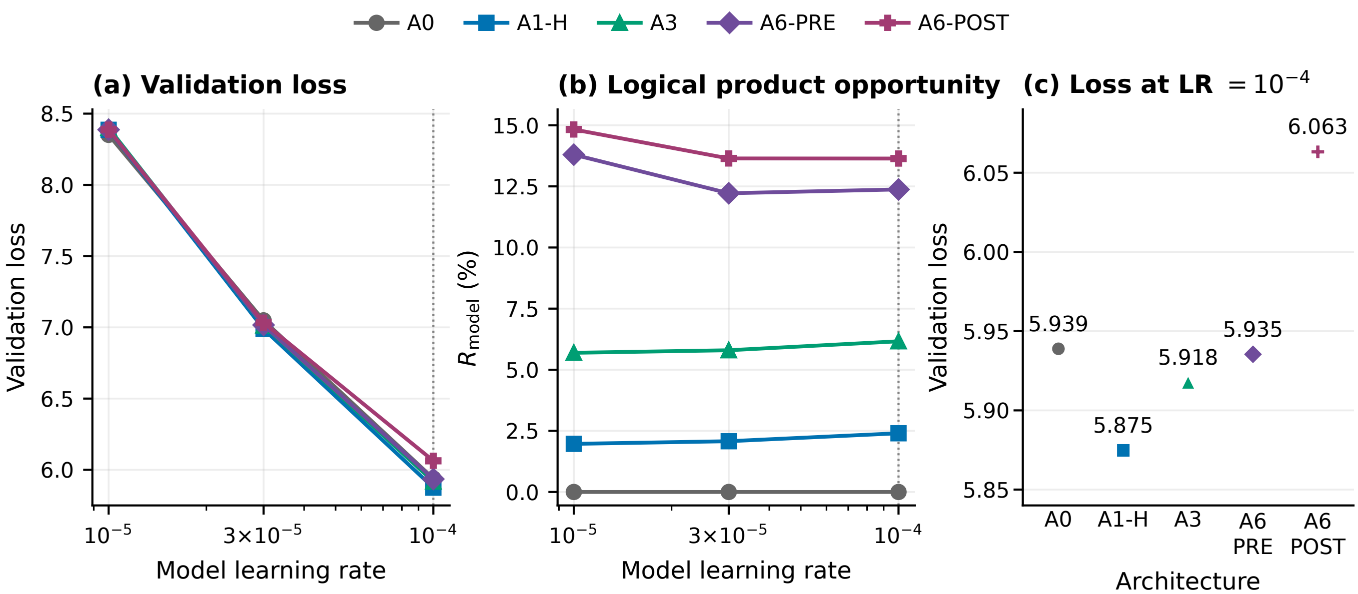

Figure 2: B0 learning-rate response for the five registered anchor architectures. All cells use AdamW without pressure. Panel (a) shows a qualitatively shared monotone endpoint-loss response across the three tested rates; panel (b) shows logical opportunity; panel (c) resolves the 10^{-4} architecture endpoints.

Finding. Relative to 3×10^{-5}, 10^{-5} raises loss by 1.3019–1.3978 for every anchor, while 10^{-4} lowers it by 0.9693–1.1140. At the common LR, A4-V versus A3 adds 3.7030 R_m pp for +0.00305 loss; adding V to the matched A5 PRE/POST parents adds 3.7735/3.7749 pp for +0.00059/+0.00184 loss. A6 PRE→POST adds 1.4238 pp for +0.01603 loss.

Limitation. The LR result ranks step-2,048 endpoints only; it does not identify a long-budget optimum. Topology comparisons expose logical reach, not realized execution speed.

3.2 B1: Fixed Thresholds

Question and setup. B1 studies two hard gate families,

$$G_\kappa^+(x) = x\mathbf{1}[x \geq \kappa], \qquad G_\kappa^\pm(x) = x\mathbf{1}[|x| \geq \kappa].$$

The complete ladders are A5-QK-PRE, A5-QK-POST, A6-PRE-QKV, and A6-POST-QKV with both gates and $\kappa \in \{0.03, 0.10, 0.30\}$. A4-Q/K/V isolate one projected tensor at $\kappa = 0.10$. A1-H, A3, and A6-POST-ALL apply G_κ^+

to every active gate at $\kappa \in \{0.03, 0.10\}$; A6-POST-ALL therefore thresholds $a/m/h/Q/K/V$, whereas A6-POST-QKV thresholds only Q/K/V. Unlisted gates remain ordinary ReLUs. G_0^+ is ordinary ReLU; $G_0^\pm$ is identity, so the latter uses A3 as its registered zero-threshold control.

Table 4: Complete B1 endpoints with the two gate families in parallel column groups. Each group reports config, validation loss, and R_m (%). The 36 positive-threshold cells are shown with their registered $\kappa = 0$ controls; repeated config IDs are shared controls. A6-POST-QKV thresholds Q/K/V, whereas A6-POST-ALL thresholds a/m/h/Q/K/V. G_0^+ is ordinary ReLU; $G_0^\pm$ is identity and uses A3 (config 125). A dash means that the gate family was not executed for that row. All cells use seed-0 AdamW at 3×10^{-5} without pressure.

Architecture	κ	G^+			$G^\pm$		
		Cfg	L	R_m (%)	Cfg	L	R_m (%)
A1-H	0	124	6.98875	2.08		–	
	0.03	204	6.98927	2.53		–	
	0.10	205	6.99409	3.35		–	
A3	0	125	7.01310	5.80		–	
	0.03	206	7.00780	6.61		–	
	0.10	207	7.02685	7.93		–	
A4-Q	0	129	7.02126	7.71	125	7.01310	5.80
	0.10	197	7.02146	8.16	200	7.01355	6.49
A4-K	0	130	7.02441	8.28	125	7.01310	5.80
	0.10	198	7.02413	9.06	201	7.01406	7.25
A4-V	0	131	7.01615	9.50	125	7.01310	5.80
	0.10	199	7.01547	11.47	202	7.01501	9.92
A5-QK-PRE	0	132	7.01586	8.45	125	7.01310	5.80
	0.03	149	7.01612	8.71	173	7.01319	6.56
	0.10	151	7.01905	9.88	175	7.01434	7.60
	0.30	153	7.03079	13.66	177	7.02590	12.00
A5-QK-POST	0	133	7.03064	9.87	125	7.01310	5.80
	0.03	161	7.03029	10.15	185	7.01265	6.62
	0.10	163	7.03090	11.10	187	7.01402	7.83
	0.30	165	7.03559	13.77	189	7.02492	12.13
A6-PRE-QKV	0	126	7.01645	12.22	125	7.01310	5.80
	0.03	155	7.01712	13.24	179	7.01432	7.90
	0.10	157	7.02195	15.70	181	7.01775	11.65
	0.30	159	7.03369	21.81	183	7.02352	19.14
A6-POST-QKV	0	127	7.03248	13.64	125	7.01310	5.80
	0.03	167	7.03290	14.59	191	7.01313	7.98
	0.10	169	7.03408	16.91	193	7.01688	11.87
	0.30	171	7.03405	21.94	195	7.02043	19.51
A6-POST-ALL	0	127	7.03248	13.64		–	
	0.03	208	7.02389	15.44		–	
	0.10	209	7.05266	18.61		–	

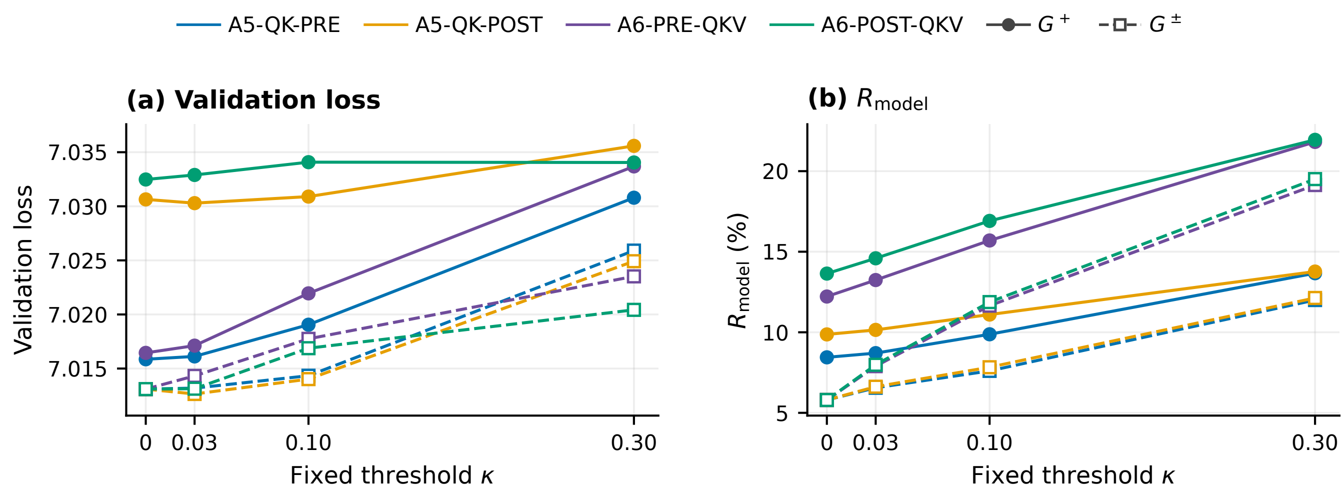

Figure 3: B1 threshold effect for the four complete A5/A6 ladders. Both panels show absolute endpoints across κ: validation loss and R_m. Colors identify architecture; solid circles and dashed open squares identify G^+ and $G^\pm$. At $\kappa = 0$, G^+ uses the same-architecture ordinary-ReLU control and $G^\pm$ uses the shared A3 identity control.

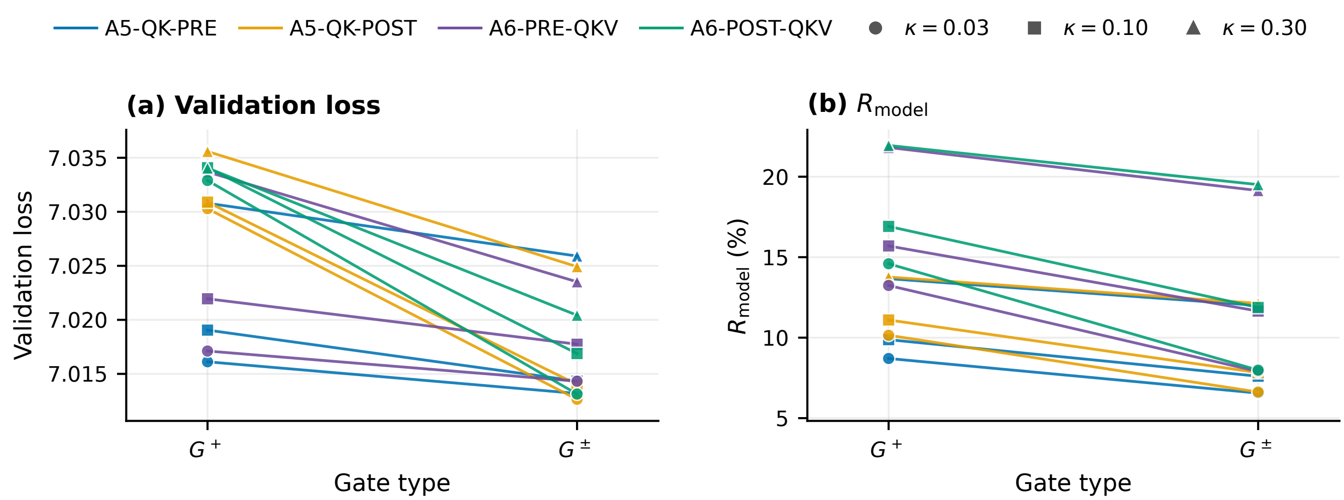

Figure 4: B1 gate-type effect at matched positive thresholds. Each segment connects the absolute G^+ and $G^\pm$ endpoints for one architecture and κ; panels show validation loss and R_m, not paired differences. Color identifies architecture and marker shape identifies κ.

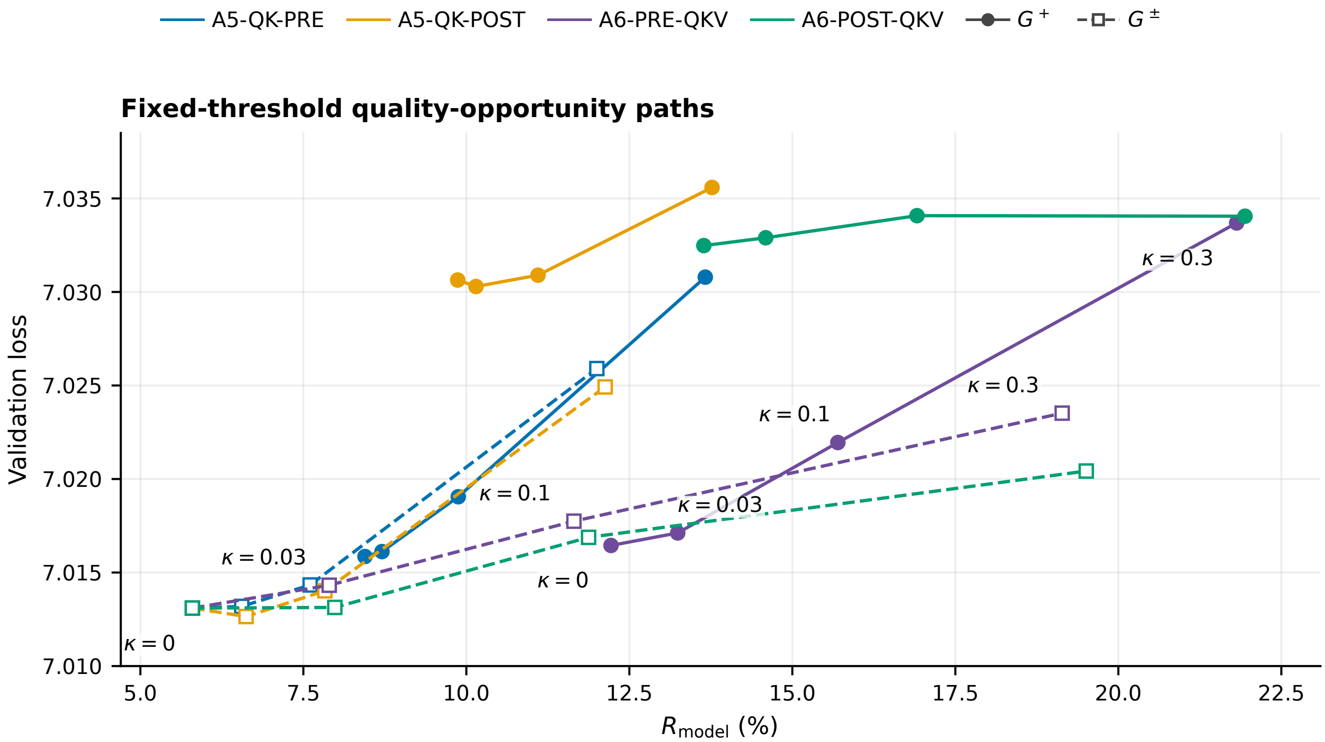

Figure 5: B1 architecture-conditioned quality–opportunity frontiers. Each path follows the tested κ ladder from its registered zero-threshold control; lower loss and larger R_m are favorable. Color identifies architecture and line/marker style identifies gate family. These are observed threshold paths, not a cross-architecture ranking.

Finding. In the four complete ladders, raising κ from 0.03 to 0.30 increases R_m in all eight architecture/gate combinations by 3.62–11.53 pp, with +0.00115–0.01657 loss. At each positive threshold, $G^{\pm}$ has lower loss than G^{+} in all 12 matched architecture–κ pairs, but also 1.64–6.61 pp lower R_m. Table 4 retains the A1/A3/A4 and A6-POST-ALL endpoints without collapsing them into paired ranges.

Limitation. Absolute κ acts on heterogeneous activation scales, especially in the all-active rows; topology and scale are not separated. The mostly one-seed screen has no prespecified equivalence margin.

3.3 B2: Learned Thresholds

Question and setup. B2 asks whether a learned positive threshold improves on fixed $\kappa = 0.10$, and whether the threshold should use raw or RMS-normalized activation magnitudes. The learned threshold is

$$\kappa = \operatorname{softplus}(\rho), \qquad s(x) = \begin{cases} x, & G^{+}, \\ |x|, & G^{\pm}. \end{cases}$$

Softplus keeps $\kappa > 0$. For a gate tensor with N elements, its root-mean-square magnitude and comparison score are

$$r(x) = \max\left(\sqrt{\frac{1}{N}\sum_i x_i^2}, 10^{-8}\right), \qquad u(x) = \begin{cases} s(x), & \text{absolute}, \\ s(x)/r(x), & \text{RMS-relative}. \end{cases}$$

Thus absolute thresholds use activation units, whereas RMS-relative thresholds scale with the current full gate tensor. The forward pass remains a hard gate:

$$y = x\,\mathbf{1}[u(x) \geq \kappa].$$

Masked values are therefore exact zeros. During training only, the derivative of the threshold comparison is replaced by the sigmoid surrogate $\operatorname{sigmoid}((u(x) - \kappa)/\tau)$. The score and RMS are treated as constants in that surrogate, so it updates ρ only; the input keeps the hard-mask zero-or-one derivative.

All cells initialize $\kappa = 0.10$, use $\tau = 0.03$, and train thresholds at 3×10^{-4} without weight decay; the model uses seed-0 AdamW at 3×10^{-5} for 2,048 steps without activation pressure. "Model-wide" means one κ for every learned gate, "per site" means one for each of Q/K/V shared across layers, and "per layer/site" means a separate κ for every layer and gate site. A6-POST-QKV learns only Q/K/V while retaining ordinary ReLUs at a/m/h; A6-POST-ALL learns a/m/h/Q/K/V.

Table 5: Complete B2 endpoints with the two gate families in parallel column groups. Each group reports config, validation loss, and R_m (%). Rows specify architecture, score scale, and threshold parameterization. A6-POST-QKV learns thresholds only at Q/K/V; A6-POST-ALL learns them at a/m/h/Q/K/V. A dash means that the gate family was not executed for that row. All 26 cells use seed-0 AdamW at 3×10^{-5} for 2,048 steps without activation pressure.

Architecture	Score scale	κ parameters	G^+			$G^\pm$		
			Cfg	L	R_m (%)	Cfg	L	R_m (%)
A1-H	Absolute	Per layer/site	237	7.00088	3.66	–		
	RMS-relative	Per layer/site	238	6.99136	2.65	–		
A3	Absolute	Per layer/site	239	7.04232	8.14	–		
	RMS-relative	Per layer/site	240	7.01449	6.90	–		
A5-QK-PRE	Absolute	Per layer/site	221	7.01955	10.54	222	7.01458	8.29
	RMS-relative	Per layer/site	229	7.01594	9.56	230	7.01283	8.17
A5-QK-POST	Absolute	Per layer/site	225	7.03098	11.63	226	7.01430	8.61
	RMS-relative	Per layer/site	233	7.03041	10.69	234	7.01226	8.49
A6-PRE-QKV	Absolute	Per layer/site	223	7.02197	16.54	224	7.01763	12.71
	RMS-relative	Per layer/site	231	7.01330	13.95	232	7.01309	9.06
A6-POST-QKV	Absolute	Model-wide	243	7.03386	17.76	245	7.01623	13.38
	Absolute	Per site	244	7.03378	18.14	246	7.01657	13.78
	Absolute	Per layer/site	227	7.03411	17.67	228	7.01617	13.09
	RMS-relative	Per layer/site	235	7.02936	14.96	236	7.01146	9.41
A6-POST-ALL	Absolute	Per layer/site	241	7.06515	19.49	–		
	RMS-relative	Per layer/site	242	7.03240	16.26	–		

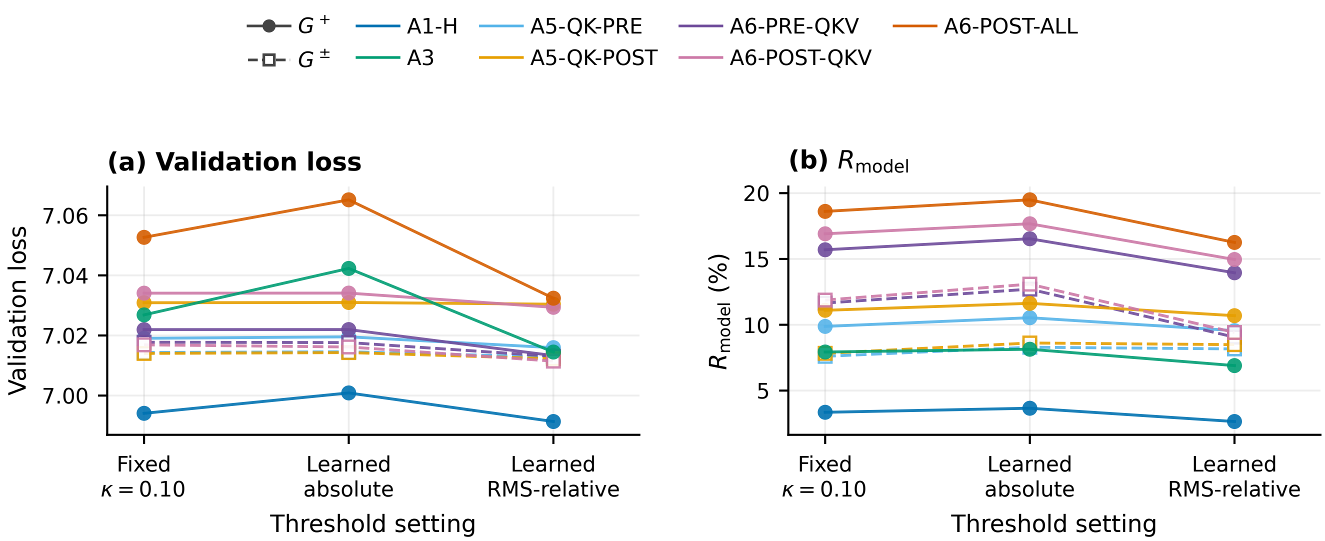

Figure 6: B2 threshold effect for the 11 architecture/gate settings with one threshold per layer/site. Each trajectory connects fixed $\kappa = 0.10$, learned absolute, and learned RMS-relative endpoints. Both panels show absolute validation loss and R_m.

Table 6: Distribution of the final learned thresholds. All cells start from $\kappa = 0.10$; values are read from the exact final checkpoint summaries. The mean pools threshold parameters: a run contributes one model-wide value, three site-shared values, or one value for every learned layer/site. Absolute and RMS-relative values live on different score scales and should not be compared as activation units.

Threshold stratum	Runs	Parameters	Min.	Mean	Max.
Q/K(/V), absolute, per layer/site	8	120	0.1001	0.1444	0.2020
Q/K(/V), RMS-relative, per layer/site	8	120	0.1074	0.1581	0.2178
a/m/h, absolute, per layer/site	3	60	0.0839	0.1318	0.1997
a/m/h, RMS-relative, per layer/site	3	60	0.0942	0.1543	0.2143
A6-POST Q/K/V, absolute, model-wide	2	2	0.1328	0.1329	0.1329
A6-POST Q/K/V, absolute, per site	2	6	0.1306	0.1552	0.1924

Finding. In the 11 per-layer/site comparisons, learning an absolute threshold raises R_m by 0.21–1.21 pp relative to fixed $\kappa = 0.10$, while loss moves by -0.00071 to +0.01546. RMS-relative scoring lowers loss by 0.00057–0.03275 and R_m by 0.13–3.67 pp relative to learned absolute scoring in all 11 comparisons. For A6-POST-QKV with absolute scoring, the three threshold parameterizations span less than 0.0004 loss and 0.70 R_m pp; Table 5 retains all 26 endpoints. The exact final threshold summaries in Table 6 show nontrivial movement from the common 0.10 initialization: stratum means span 0.1318–0.1581 and individual values span 0.0839–0.2178.

Limitation. Thresholds optimize task loss, not sparsity. No fixed RMS-relative control exists, only A6-POST-QKV tests all three threshold parameterizations, and the screen has one seed.

3.4 B3: Pressure and Orthogonalization

Question and setup. For active tensors $s \in \mathcal{S}$, pressure first averages scalars within a tensor and then weights tensors equally:

$$\mathcal{L}_{\mathrm{L1}} = \frac{1}{|\mathcal{S}|}\sum_s \frac{1}{N_s}\sum_i |x_{s,i}|, \quad \mathcal{L}_{\mathrm{R}} = 1 - \frac{1}{|\mathcal{S}|}\sum_s \frac{1}{N_s}\sum_i \left(1 - \frac{x_{s,i}^2}{c^2}\right) e^{-x_{s,i}^2/(2\sigma^2)}.$$

L1N/RN add weighted pressure to the training loss. OL1/OR instead apply the pressure correction after the AdamW task step, removing the component that conflicts with that task step. Their pressure/task update ratio is capped at 0.5 as a fixed stability control. In the Ricker pressure, c is the zero-crossing scale and σ controls the envelope width. The promotion guardrail remains $L_{\text{method}} - L_{\text{same-architecture AdamW}} \leq 0.05$; it is an evaluation rule, not a training objective.

Pressure weight and orthogonalization. B3 crosses A3 and A6-POST with L1 weights $\{0.15, 1, 5\}$ and Ricker weights $\{0.1, 0.3, 1\}$, fixing $c = \sigma = 0.10$ for the Ricker weight sweep.

Table 7: Complete B3 pressure-weight endpoints. Each method group reports config, validation loss, and R_m (%). L1 rows compare L1N with OL1; Ricker rows compare RN with OR at $c = \sigma = 0.10$. AdamW is the matched same-architecture control. All cells use seed 0 and 2,048 steps.

Architecture	Pressure	w	AdamW			Naive			Orthogonal		
			Cfg	L	R_m (%)	Cfg	L	R_m (%)	Cfg	L	R_m (%)
A3	L1	0.15	125	7.01310	5.80	257	7.02421	5.07	258	7.01421	5.13
		1	125	7.01310	5.80	248	7.05338	4.59	249	7.02002	4.69
		5	125	7.01310	5.80	261	7.17578	5.18	262	7.02909	4.71
	Ricker	0.1	125	7.01310	5.80	266	7.03683	7.23	267	7.03121	7.28
		0.3	125	7.01310	5.80	252	7.10085	8.16	253	7.04616	7.87
		1	125	7.01310	5.80	270	7.20123	8.77	271	7.05883	8.25
A6-POST	L1	0.15	127	7.03248	13.64	259	7.04048	18.77	260	7.03898	17.70
		1	127	7.03248	13.64	250	7.05312	19.60	251	7.03895	17.69
		5	127	7.03248	13.64	263	7.09562	21.32	264	7.03870	17.70
	Ricker	0.1	127	7.03248	13.64	268	7.05868	21.88	269	7.03926	17.55
		0.3	127	7.03248	13.64	254	7.09617	23.67	255	7.03930	17.56
		1	127	7.03248	13.64	272	7.16248	24.90	273	7.03909	17.56

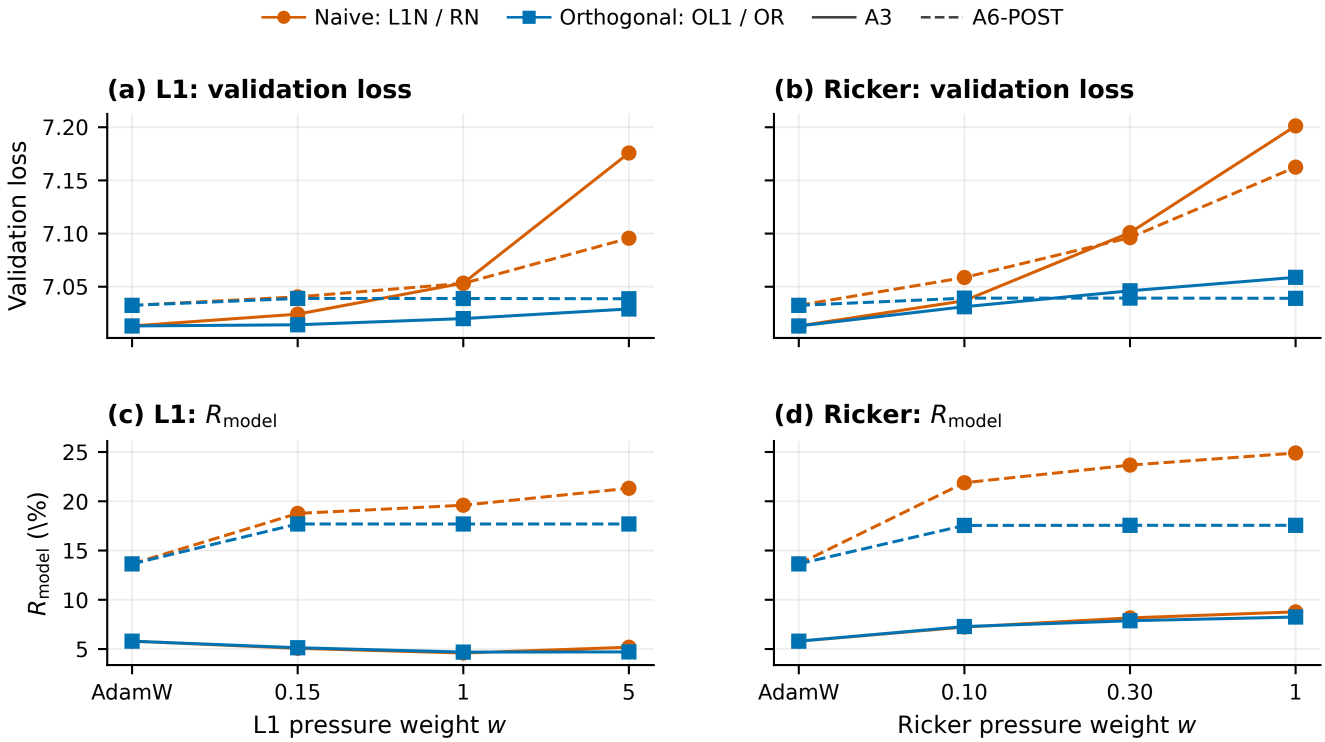

Figure 7: B3 pressure-weight response. Each trajectory starts at the matched AdamW control and continues through the three pressure weights. Panels show absolute validation loss and R_m. Color and marker identify naive versus orthogonal pressure; line style identifies A3 versus A6-POST. L1 and Ricker weights are separate axes because their objective scales differ.

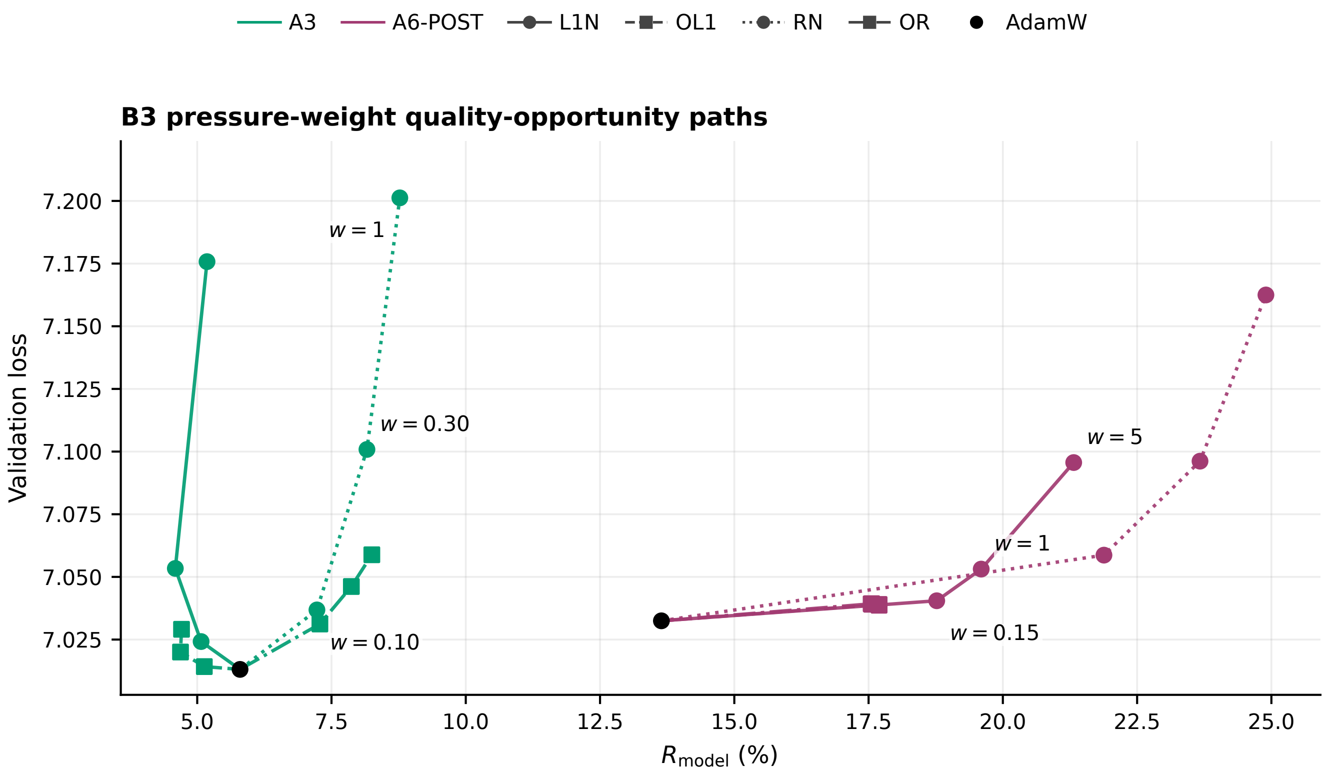

Figure 8: B3 pressure-weight quality–opportunity frontiers. Each path begins at its matched AdamW control and then follows one L1N, OL1, RN, or OR weight ladder; lower loss and larger R_m are favorable. L1 and Ricker weights index different objectives and are not numerically interchangeable.

Finding. The orthogonal method has lower loss than its matched naive method in all 12 weight comparisons, but it does not preserve the full naive R_m gain. On A6-POST, OL1 changes by less than 0.0003 loss and 0.002 R_m pp across its weight ladder; OR changes by less than 0.0003 loss and 0.011 pp. The corresponding naive ladders remain weight-sensitive. This flat A6 orthogonal response coincides with the update-ratio cap binding on every logged orthogonal update.

Ricker geometry. The geometry screen fixes $w = 0.30$. One sweep couples $c = \sigma \in \{0.05, 0.10, 0.50\}$; the other fixes $c = 0.10$ and varies $\sigma \in \{0.05, 0.10, 0.20\}$.

Table 8: Complete B3 Ricker-geometry endpoints at $w = 0.30$. Each method group reports config, validation loss, and R_m (%). The basin sweep couples $c = \sigma$; the shape sweep fixes $c = 0.10$ and varies σ. AdamW is the matched same-architecture control.

Architecture	Sweep	Value	AdamW			RN			OR		
			Cfg	L	R_m (%)	Cfg	L	R_m (%)	Cfg	L	R_m (%)
A3	$c = \sigma$	0.05	125	7.01310	5.80	275	7.12042	8.36	276	7.03339	7.84
		0.1	125	7.01310	5.80	252	7.10085	8.16	253	7.04616	7.87
		0.5	125	7.01310	5.80	279	7.07964	7.32	280	7.05550	7.33
	σ at $c = 0.10$	0.05	125	7.01310	5.80	284	7.10143	8.68	285	7.05176	8.24
		0.1	125	7.01310	5.80	252	7.10085	8.16	253	7.04616	7.87
		0.2	125	7.01310	5.80	288	7.12562	8.09	289	7.04407	7.60
A6-POST	$c = \sigma$	0.05	127	7.03248	13.64	277	7.10916	23.23	278	7.03514	17.02
		0.1	127	7.03248	13.64	254	7.09617	23.67	255	7.03930	17.56
		0.5	127	7.03248	13.64	281	7.07163	21.01	282	7.04037	18.10
	σ at $c = 0.10$	0.05	127	7.03248	13.64	286	7.09446	24.23	287	7.03708	17.39
		0.1	127	7.03248	13.64	254	7.09617	23.67	255	7.03930	17.56
		0.2	127	7.03248	13.64	290	7.11203	24.03	291	7.04009	17.85

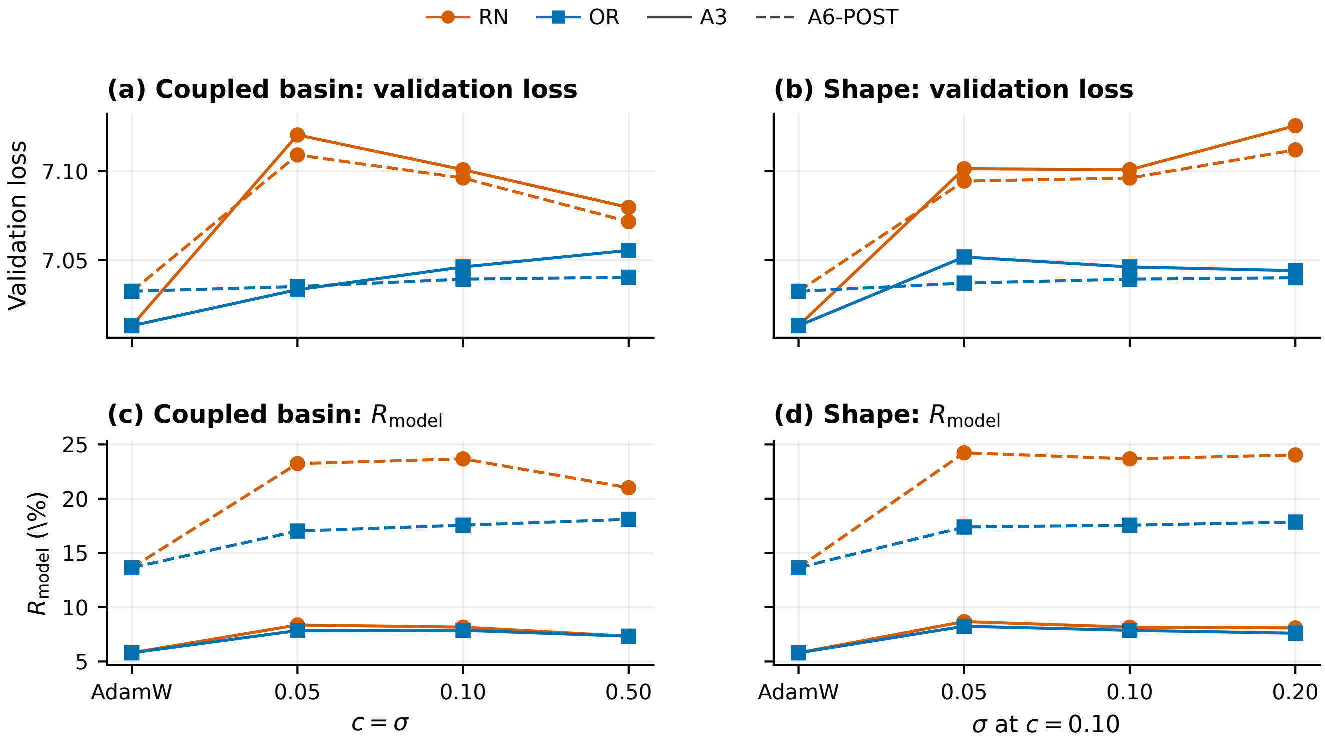

Figure 9: B3 Ricker-geometry response at $w = 0.30$. Each trajectory starts at the matched AdamW control; the remaining points vary only the displayed geometry parameter. Panels show absolute validation loss and R_m. Color and marker identify RN versus OR; line style identifies architecture.

Finding. Geometry interacts with both update rule and architecture. Increasing the coupled basin from 0.05 to 0.50 lowers RN loss by 0.0408 on A3 and 0.0375 on A6-POST, while reducing R_m by 1.04 and 2.22 pp. For OR, the same change raises A6-POST R_m by 1.08 pp but lowers A3 R_m by 0.51 pp. The shape sweep is also non-monotone for RN. No single Ricker geometry dominates across method and architecture.

Limitation. The screen has one seed and 2,048 steps. Tensors are weighted equally rather than by width or matmul cost, so pressure does not directly optimize R_m. The cap also binds on 52.38% of logged A3 central updates and every logged A6-POST orthogonal update; these cells therefore characterize the budgeted procedure, not unconstrained orthogonal geometry.

3.5 B4: Seed Sensitivity

Question and setup. B4 repeats ten sentinels—A0, A1-H, A3, A6-PRE, A6-POST, fixed G^+, fixed $G^\pm$, learned $G^\pm$, L1N, and OR—while jointly changing model-initialization and data-order seeds from 0/0 to 1/1. Every other scientific field is fixed.

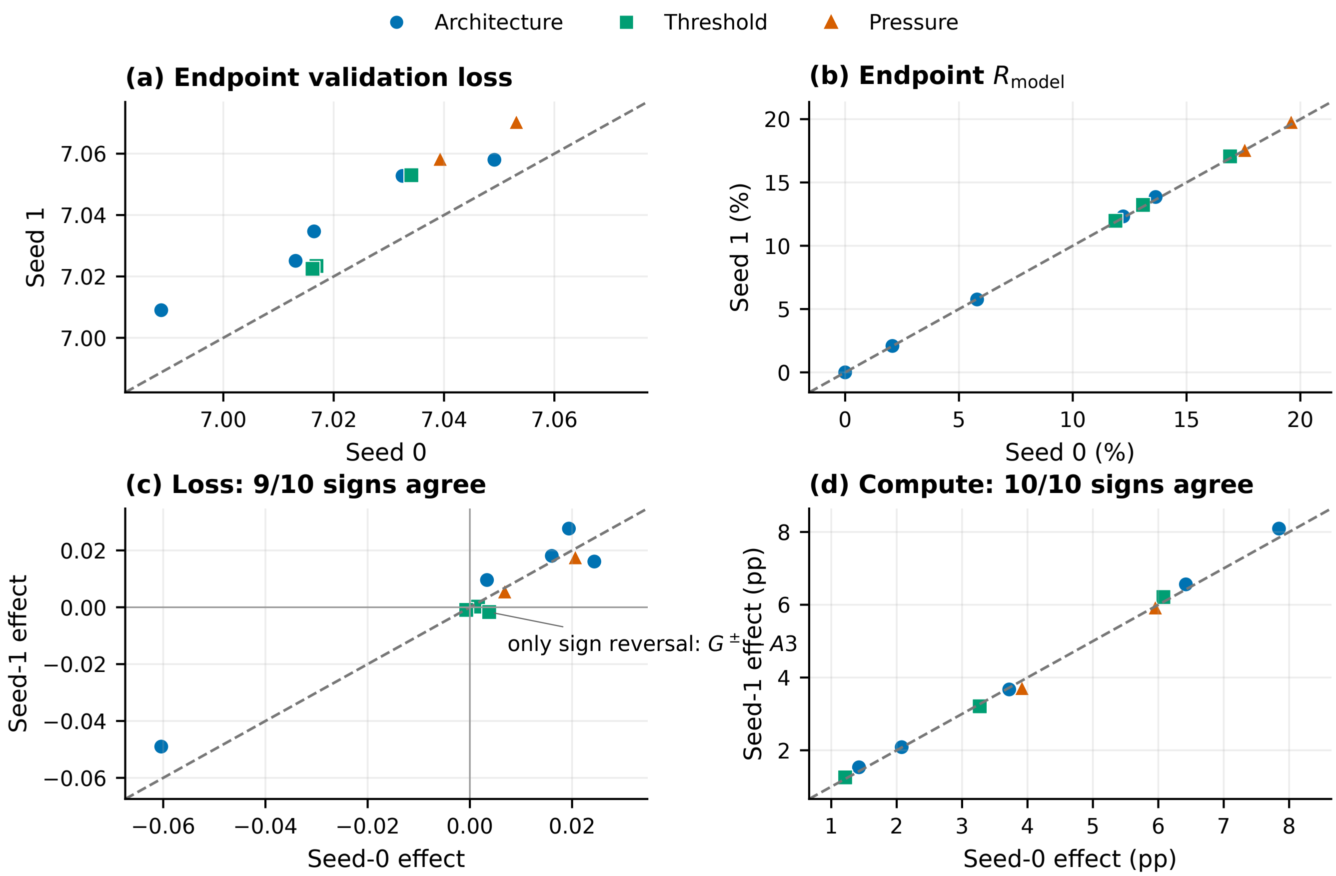

Figure 10: B4 seed sensitivity. Top: absolute seed-0/seed-1 endpoints. Bottom: within-seed matched effects. Dashed lines are identity. The only loss-sign reversal is labelled. The contrasts share controls and were selected after the seed-0 screen.

Finding. All ten R_m effect signs and nine of ten loss-effect signs agree. Seed-1 endpoint losses shift by +0.00631–0.02028; absolute R_m shifts are at most 0.2094 pp. The sole reversal is fixed $G^{\pm}_{0.10}$ minus A3.

Limitation. This is a two-point coupled sensitivity check, not a variance estimate or independent confirmation. Initialization and data-order effects cannot be separated, and B4 has no seed-1 RN or OL1 cell.

4 Synthesis, Descriptive Frontier, and Next Step

For descriptive context, the common-condition landscape retains every valid seed-0 cell at model LR 3×10^{-5}, the shared 2,048-step budget, and the frozen selection set. This excludes ten B0 LR flanks and ten B4 seed-1 sentinels, leaving $n = 112$. Point j is nondominated when no eligible point i satisfies

$$L_i \leq L_j, \qquad R_{m,i} \geq R_{m,j},$$

with at least one strict inequality. This is an exact descriptive envelope, not the promoted experimental panel.

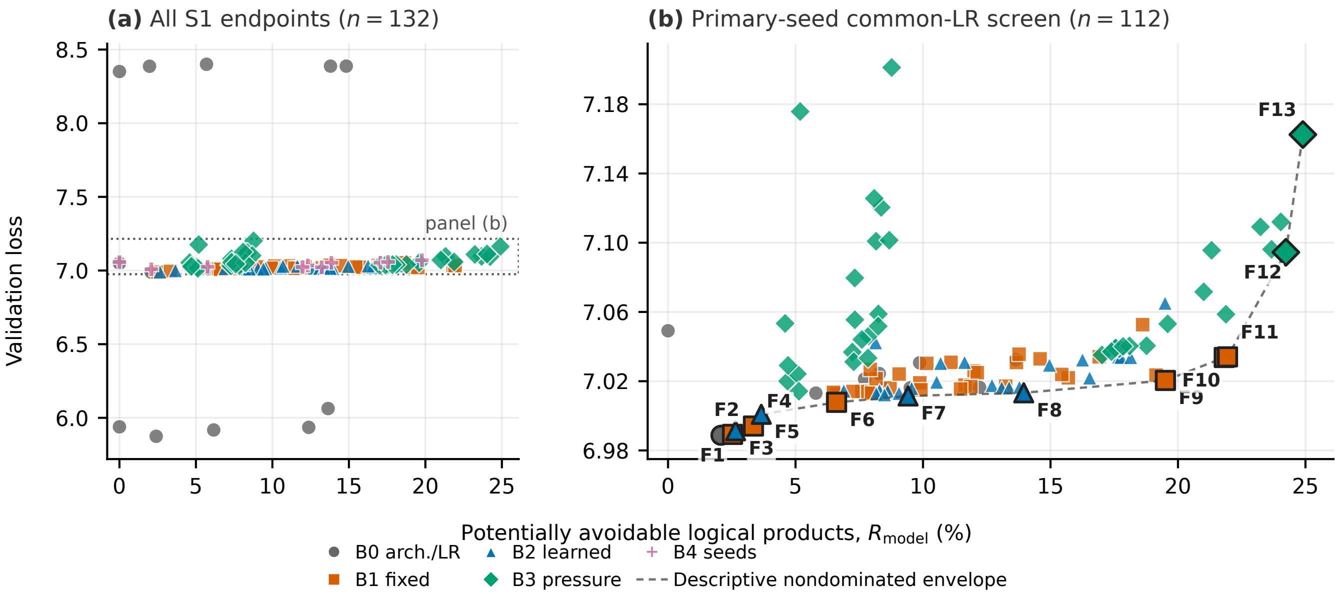

Figure 11: S1 quality–compute landscape. Panel (a) shows all 132 scientific endpoints and therefore mixes learning rates and seeds. Panel (b) restricts to the primary-seed common-LR cohort and labels the exact nondominated envelope F1–F13. Architecture ceilings differ, so cross-topology R_m must be read against gate scope and matched controls.

Table 9: Exact members of the descriptive nondominated envelope in Figure 11. The eligible cohort fixes model/data-order seeds at 0/0, model LR at 3×10^{-5}, the 2,048-step budget, and the frozen selection partition ($n = 112$). F12–F13 exceed the predeclared +0.05 pressure-loss guardrail and are descriptive, not promotion choices.

ID	Cfg	Block	Design	L	R_m (%)
F1	124	B0	A1-H AdamW	6.98875	2.08
F2	204	B1	A1-H G^+, $\kappa = 0.03$	6.98927	2.53
F3	238	B2	A1-H ATG G^+, RMS, PLS	6.99136	2.65
F4	205	B1	A1-H G^+, $\kappa = 0.10$	6.99409	3.35
F5	237	B2	A1-H ATG G^+, ABS, PLS	7.00088	3.66
F6	206	B1	A3 G^+, $\kappa = 0.03$	7.00780	6.61
F7	236	B2	A6-POST ATG $G^{\pm}$, RMS, PLS	7.01146	9.41
F8	231	B2	A6-PRE ATG G^+, RMS, PLS	7.01330	13.95
F9	195	B1	A6-POST $G^{\pm}$, $\kappa = 0.30$	7.02043	19.51
F10	159	B1	A6-PRE G^+, $\kappa = 0.30$	7.03369	21.81
F11	171	B1	A6-POST G^+, $\kappa = 0.30$	7.03405	21.94
F12	286	B3	A6-POST RN, $(w, c, \sigma) = (0.3, 0.1, 0.05)$	7.09446	24.23
F13	272	B3	A6-POST RN, $(w, c, \sigma) = (1, 0.1, 0.1)$	7.16248	24.90

Three conclusions survive the blockwise controls:

- **Architecture precedes optimization.** Gate placement defines which operands can be zero. V and POST Q/K enlarge observed opportunity, while projections, attention aggregation, and residual addition regenerate dense values.
- **Thresholds and pressure are different controls.** Fixed and learned gates set forward support directly. Pressure reshapes activation distributions, and orthogonalization changes how much task damage the pressure update accepts. Their weights and endpoints are not interchangeable.
- **The global envelope is descriptive.** F12–F13 obtain the largest R_m through RN but violate the predeclared +0.05 pressure-loss guardrail. Matched controls required for causal comparison need not lie on the envelope.

The next rung should freeze a matched panel before observing new data: A0, A3, and ordinary A6-PRE/A6-POST controls; fixed POST-QKV $G^+_{0.10}$ and $G^{\pm}_{0.10}$, plus high-opportunity $G^+_{0.30}$; learned absolute $G^{\pm}$ with one threshold per layer/site; and matched L1N/OL1 and RN/OR anchors. First extend the Pythia-14M token budget under the frozen ladder: S2 uses seeds 0/1 on the selection partition; S3 uses fresh seeds 2–4 on the document-disjoint campaign-confirmation partition. That partition is not untouched, so claim-grade confirmation still requires a new independent evaluation source. Then scale the frozen panel across the Pythia family, recomputing topology ceilings at each size. Freeze topology and gate family, but do not assume that absolute κ or Ricker (c, σ) transfers across activation scales; predeclare a normalized transfer or report frozen and retuned curves separately. Sparse-kernel latency, energy, masking, indexing, RoPE, softmax, and residual overhead should be measured only after the logical frontier is stable.

5 Reproducibility and Provenance

The data-derived figures, tables, and provenance sidecar are regenerated from the applied config/run registries by:

```
make plot-report07
```

On the recorded Windows host, the directly executable equivalent is `mingw32-make plot-report07`. The canonical PDF is then compiled by running `pdflatex 07-2026-07-27-s1-ablation-study.tex` twice from the report directory. The plotting target generates the topology atlas, Figures 103–110, 112–114, 116, and 118, the frontier table, the 132-row appendix, PNG previews, and the provenance sidecar. The registry-driven generators validate the expected 20/36/26/40/10 block counts and the 311,296-position propagation artifacts; the topology atlas is derived from the frozen architecture specification.

Model revision: `7386d9a4ae45aef494a6e704910394def3037fc5`.
MiniPile revision: `18ad1b0c701eaa0de03d3cecfdd769cbc70ffbd0`.
Frozen selection hash: `ffc857a6f0771929dd75c93bc17729de98a692f3a175ac5742cc9d101ff4ea47`. Execution used Windows 11, Python 3.12.10, PyTorch 2.11.0+cu128, CUDA 12.8, and driver 572.84.

Durable scientific sources are `docs/experimental-design/config-registry.yaml`, `docs/experimental-design/run-registry.yaml`, the B0–B4 result documents in `docs/experimental-design/`, and each canonical propagation artifact. Registry closure is recorded at `5ddc0ca1e650f90c1035e14a3b7d6f69384d0205`. The canonical B4 diagnostic is config 303, run `001-20260727-190455-ba5f3286`; its `activation_propagation.json` SHA-256 is `2f2c24b31f953f58e1406aa2013992bff5526713f848ccc0c5da9e8c6393b4d5`.

A Canonical S1 Core Endpoint Tables

The following tables enumerate all 132 scientific cells. Values come directly from canonical run-registry endpoints. z_Q^g, z_K^g, z_V^g are explicit gate-output zero rates, not post-RoPE QK operands. A dash means that a gate or measurement is not applicable. Downstream $z_Q^{QK}, z_K^{QK}, z_V^{PV}, z_C^{W_o}$ and per-operation product counters remain in the machine-readable registry/raw artifacts rather than this compact core table. Engineering and diagnostic configs are intentionally excluded.

A.1 B0: architecture and learning-rate endpoints

Table 10: Canonical S1-B0 core scientific endpoints. R_m and all z columns are percentages; z_Q^g, z_K^g, z_V^g are gate-output zeros.

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z_Q^g	z_K^g	z_V^g
123	A0	AdamW; lr=3×10^{-5}	0/0	7.04913	< 0.0001	–	–	–	–	–	–
124	A1-H; G^+ h	AdamW; lr=3×10^{-5}; κ=0	0/0	6.98875	2.077	–	–	48.543	–	–	–
125	A3; G^+ a/m/h	AdamW; lr=3×10^{-5}; κ=0	0/0	7.01310	5.797	51.121	51.179	46.001	–	–	–
126	A6-PRE; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0; PRE	0/0	7.01645	12.219	51.002	51.148	45.867	15.955	28.680	43.460
127	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0; POST	0/0	7.03248	13.642	50.996	51.090	45.919	22.610	34.973	43.614
129	A4-Q; G^+ a/m/h/Q	AdamW; lr=3×10^{-5}; κ=0; POST	0/0	7.02126	7.708	51.131	51.173	45.966	22.694	–	–
130	A4-K; G^+ a/m/h/K	AdamW; lr=3×10^{-5}; κ=0; POST	0/0	7.02441	8.285	51.119	51.149	45.897	–	29.194	–
131	A4-V; G^+ a/m/h/V	AdamW; lr=3×10^{-5}; κ=0	0/0	7.01615	9.500	51.048	51.199	45.956	–	–	43.206
132	A5-QK-PRE; G^+ a/m/h/Q/K	AdamW; lr=3×10^{-5}; κ=0; PRE	0/0	7.01586	8.445	51.070	51.119	45.996	15.592	27.218	–
133	A5-QK-POST; G^+ a/m/h/Q/K	AdamW; lr=3×10^{-5}; κ=0; POST	0/0	7.03064	9.868	51.113	51.135	45.960	23.098	34.280	–
135	A0	AdamW; lr=1×10^{-5}	0/0	8.35104	< 0.0001	–	–	–	–	–	–
136	A1-H; G^+ h	AdamW; lr=1×10^{-5}; κ=0	0/0	8.38652	1.968	–	–	46.013	–	–	–
137	A3; G^+ a/m/h	AdamW; lr=1×10^{-5}; κ=0	0/0	8.40015	5.693	50.490	50.553	44.666	–	–	–
138	A6-PRE; G^+ a/m/h/Q/K/V	AdamW; lr=1×10^{-5}; κ=0; PRE	0/0	8.38739	13.798	50.692	50.780	44.462	33.861	41.380	43.887
139	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=1×10^{-5}; κ=0; POST	0/0	8.38743	14.827	50.690	50.785	44.455	36.139	44.505	43.929
140	A0	AdamW; lr=1×10^{-4}	0/0	5.93887	< 0.0001	–	–	–	–	–	–
141	A1-H; G^+ h	AdamW; lr=1×10^{-4}; κ=0	0/0	5.87474	2.402	–	–	56.144	–	–	–
142	A3; G^+ a/m/h	AdamW; lr=1×10^{-4}; κ=0	0/0	5.91768	6.164	50.906	51.237	54.670	–	–	–
143	A6-PRE; G^+ a/m/h/Q/K/V	AdamW; lr=1×10^{-4}; κ=0; PRE	0/0	5.93539	12.375	50.734	51.314	55.088	11.062	17.800	52.222
144	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=1×10^{-4}; κ=0; POST	0/0	6.06320	13.639	50.396	51.055	54.586	17.121	24.023	52.621

A.2 B1: fixed-threshold endpoints

Table 11: Canonical S1-B1 core scientific endpoints. R_m and all z columns are percentages; z_Q^g, z_K^g, z_V^g are gate-output zeros.

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z_Q^g	z_K^g	z_V^g
149	A5-QK-PRE; ReLU a/m/h; G^+ Q/K	AdamW; lr=3×10^{-5}; κ=0.03; PRE; qk	0/0	7.01612	8.706	51.068	51.118	45.955	18.626	30.638	–
151	A5-QK-PRE; ReLU a/m/h; G^+ Q/K	AdamW; lr=3×10^{-5}; κ=0.1; PRE; qk	0/0	7.01905	9.876	51.057	51.094	45.949	32.217	44.069	–
153	A5-QK-PRE; ReLU a/m/h; G^+ Q/K	AdamW; lr=3×10^{-5}; κ=0.3; PRE; qk	0/0	7.03079	13.665	51.069	51.068	45.909	87.580	90.006	–

Continued on next page

Table 11 (continued)

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z^g_Q	z^g_K	z^g_V
155	A6-PRE; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.03; PRE; qkv	0/0	7.01712	13.242	50.991	51.138	45.974	19.742	32.552	50.546
157	A6-PRE; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; PRE; qkv	0/0	7.02195	15.698	50.848	51.030	46.073	34.100	44.857	66.315
159	A6-PRE; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.3; PRE; qkv	0/0	7.03369	21.815	50.728	50.928	46.211	87.044	90.638	90.727
161	A5-QK-POST; ReLU a/m/h; G$^+$ Q/K	AdamW; lr=3×10^{-5}; κ=0.03; POST; qk	0/0	7.03029	10.148	51.133	51.152	45.974	25.501	37.782	–
163	A5-QK-POST; ReLU a/m/h; G$^+$ Q/K	AdamW; lr=3×10^{-5}; κ=0.1; POST; qk	0/0	7.03090	11.099	51.117	51.143	45.932	36.316	49.138	–
165	A5-QK-POST; ReLU a/m/h; G$^+$ Q/K	AdamW; lr=3×10^{-5}; κ=0.3; POST; qk	0/0	7.03559	13.768	51.101	51.096	45.936	84.438	87.556	–
167	A6-POST; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.03; POST; qkv	0/0	7.03290	14.592	51.020	51.113	45.863	25.246	39.159	50.716
169	A6-POST; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; POST; qkv	0/0	7.03408	16.912	50.964	51.083	46.092	37.115	50.232	66.158
171	A6-POST; ReLU a/m/h; G$^+$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.3; POST; qkv	0/0	7.03405	21.943	50.797	50.995	46.145	84.622	88.461	90.730
173	A5-QK-PRE; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.03; PRE; qk	0/0	7.01319	6.555	51.165	51.224	45.998	2.848	7.119	–
175	A5-QK-PRE; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.1; PRE; qk	0/0	7.01434	7.604	51.171	51.229	45.975	8.939	17.907	–
177	A5-QK-PRE; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.3; PRE; qk	0/0	7.02590	12.002	51.073	51.084	45.934	62.548	67.495	–
179	A6-PRE; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.03; PRE; qkv	0/0	7.01432	7.897	51.069	51.132	45.906	2.799	7.072	15.813
181	A6-PRE; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; PRE; qkv	0/0	7.01775	11.650	50.962	51.069	46.049	8.157	17.014	48.218
183	A6-PRE; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.3; PRE; qkv	0/0	7.02352	19.139	50.706	50.976	46.324	57.443	62.369	88.138
185	A5-QK-POST; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.03; POST; qk	0/0	7.01265	6.622	51.139	51.206	45.973	2.827	7.062	–
187	A5-QK-POST; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.1; POST; qk	0/0	7.01402	7.830	51.150	51.205	45.978	8.816	17.120	–
189	A5-QK-POST; ReLU a/m/h; G$^\pm$ Q/K	AdamW; lr=3×10^{-5}; κ=0.3; POST; qk	0/0	7.02492	12.129	51.087	51.113	45.968	60.123	65.152	–
191	A6-POST; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.03; POST; qkv	0/0	7.01313	7.983	50.991	51.065	45.923	2.816	7.252	15.884
193	A6-POST; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; POST; qkv	0/0	7.01688	11.875	50.949	51.058	46.006	8.078	16.576	48.377
195	A6-POST; ReLU a/m/h; G$^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.3; POST; qkv	0/0	7.02043	19.508	50.728	51.031	46.395	57.007	61.742	88.164
197	A4-Q; ReLU a/m/h; G$^+$ Q	AdamW; lr=3×10^{-5}; κ=0.1; POST; q	0/0	7.02146	8.160	51.148	51.190	45.963	28.031	–	–
198	A4-K; ReLU a/m/h; G$^+$ K	AdamW; lr=3×10^{-5}; κ=0.1; POST; k	0/0	7.02413	9.064	51.125	51.160	45.965	–	38.213	–
199	A4-V; ReLU a/m/h; G$^+$ V	AdamW; lr=3×10^{-5}; κ=0.1; v	0/0	7.01547	11.473	50.915	51.126	46.045	–	–	65.872
200	A4-Q; ReLU a/m/h; G$^\pm$ Q	AdamW; lr=3×10^{-5}; κ=0.1; POST; q	0/0	7.01355	6.487	51.117	51.187	46.008	8.065	–	–
201	A4-K; ReLU a/m/h; G$^\pm$ K	AdamW; lr=3×10^{-5}; κ=0.1; POST; k	0/0	7.01406	7.254	51.145	51.202	46.005	–	16.998	–
202	A4-V; ReLU a/m/h; G$^\pm$ V	AdamW; lr=3×10^{-5}; κ=0.1; v	0/0	7.01501	9.918	50.899	51.014	46.012	–	–	48.298
204	A1-H; G$^+$ h	AdamW; lr=3×10^{-5}; κ=0.03	0/0	6.98927	2.529	–	–	59.110	–	–	–

Continued on next page

Table 11 (continued)

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z_Q^g	z_K^g	z_V^g
205	A1-H; G^+ h	AdamW; lr=3×10^{-5}; κ=0.1	0/0	6.99409	3.350	–	–	78.310	–	–	–
206	A3; G^+ a/m/h	AdamW; lr=3×10^{-5}; κ=0.03	0/0	7.00780	6.609	52.360	52.445	62.792	–	–	–
207	A3; G^+ a/m/h	AdamW; lr=3×10^{-5}; κ=0.1	0/0	7.02685	7.931	55.064	55.148	88.948	–	–	–
208	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0.03; POST	0/0	7.02389	15.441	52.380	52.502	62.772	25.507	39.479	51.164
209	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; POST	0/0	7.05266	18.615	54.731	54.909	88.888	33.545	46.571	65.666

A.3 B2: learned-threshold endpoints

Table 12: Canonical S1-B2 core scientific endpoints. R_m and all z columns are percentages; z_Q^g, z_K^g, z_V^g are gate-output zeros.

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z_Q^g	z_K^g	z_V^g
221	A5-QK-PRE; ReLU a/m/h; ATG G^+ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; PRE	0/0	7.01955	10.535	51.091	51.127	45.941	38.994	52.627	–
222	A5-QK-PRE; ReLU a/m/h; ATG $G^\pm$ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; PRE	0/0	7.01458	8.293	51.127	51.188	46.019	12.934	26.119	–
223	A6-PRE; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; PRE	0/0	7.02197	16.536	50.861	51.041	46.046	39.894	53.159	68.868
224	A6-PRE; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; PRE	0/0	7.01763	12.706	50.993	51.103	46.028	12.627	24.215	53.228
225	A5-QK-POST; ReLU a/m/h; ATG G^+ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	0/0	7.03098	11.629	51.112	51.128	45.947	41.509	56.523	–
226	A5-QK-POST; ReLU a/m/h; ATG $G^\pm$ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	0/0	7.01430	8.615	51.122	51.182	45.947	12.873	24.547	–
227	A6-POST; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	0/0	7.03411	17.674	50.935	51.060	46.077	41.753	57.800	68.750
228	A6-POST; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	0/0	7.01617	13.087	50.991	51.104	46.004	12.539	23.639	53.324
229	A5-QK-PRE; ReLU a/m/h; ATG G^+ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; PRE	0/0	7.01594	9.556	51.080	51.127	46.003	27.984	39.265	–
230	A5-QK-PRE; ReLU a/m/h; ATG $G^\pm$ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; PRE	0/0	7.01283	8.167	51.174	51.230	46.036	11.620	22.389	–
231	A6-PRE; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; PRE	0/0	7.01330	13.951	50.967	51.120	45.989	29.080	42.468	48.279
232	A6-PRE; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; PRE	0/0	7.01309	9.059	51.116	51.183	45.940	11.005	22.101	11.261
233	A5-QK-POST; ReLU a/m/h; ATG G^+ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; POST	0/0	7.03041	10.690	51.108	51.130	45.900	32.145	43.063	–
234	A5-QK-POST; ReLU a/m/h; ATG $G^\pm$ Q/K	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; POST	0/0	7.01226	8.489	51.143	51.204	46.028	12.069	22.638	–
235	A6-POST; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; POST	0/0	7.02936	14.958	50.980	51.080	45.992	31.563	44.998	48.329
236	A6-POST; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; POST	0/0	7.01146	9.415	51.133	51.196	45.945	11.786	22.290	11.350
237	A1-H; ATG G^+ h	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site	0/0	7.00088	3.657	–	–	85.501	–	–	–

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Table 12 (continued)

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z^g_Q	z^g_K	z^g_V
238	A1-H; ATG G^+ h	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site	0/0	6.99136	2.649	–	–	61.921	–	–	–
239	A3; ATG G^+ a/m/h	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site	0/0	7.04232	8.140	54.973	56.102	92.960	–	–	–
240	A3; ATG G^+ a/m/h	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site	0/0	7.01449	6.897	55.816	58.343	61.033	–	–	–
241	A6-POST; ATG G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	0/0	7.06515	19.493	54.584	55.890	92.583	39.416	54.936	67.207
242	A6-POST; ATG G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; RMS; layer/site; POST	0/0	7.03240	16.256	56.062	58.674	60.810	32.471	46.274	49.001
243	A6-POST; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; global; POST	0/0	7.03386	17.759	50.941	51.076	46.074	40.755	54.001	72.321
244	A6-POST; ReLU a/m/h; ATG G^+ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; site; POST	0/0	7.03378	18.145	50.932	51.066	46.028	43.112	61.049	72.191
245	A6-POST; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; global; POST	0/0	7.01623	13.377	50.966	51.088	46.032	10.740	20.917	60.267
246	A6-POST; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; site; POST	0/0	7.01657	13.782	50.962	51.089	46.029	12.840	26.272	59.300

A.4 B3: pressure endpoints

Table 13: Canonical S1-B3 core scientific endpoints. R_m and all z columns are percentages; z^g_Q, z^g_K, z^g_V are gate-output zeros.

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z^g_Q	z^g_K	z^g_V
248	A3; G^+ a/m/h	L1N; κ=0; w=1	0/0	7.05338	4.592	37.328	32.600	46.754	–	–	–
249	A3; G^+ a/m/h	OL1; κ=0; w=1; budget=0.5	0/0	7.02002	4.689	40.982	32.222	46.647	–	–	–
250	A6-POST; G^+ a/m/h/Q/K/V	L1N; κ=0; w=1; POST	0/0	7.05312	19.597	38.003	30.597	42.598	89.740	89.741	82.149
251	A6-POST; G^+ a/m/h/Q/K/V	OL1; κ=0; w=1; budget=0.5; POST	0/0	7.03895	17.693	51.014	51.045	46.015	88.062	88.077	44.755
252	A3; G^+ a/m/h	RN; κ=0; w=0.3; c=0.1; σ=0.1	0/0	7.10085	8.160	68.479	68.349	71.044	–	–	–
253	A3; G^+ a/m/h	OR; κ=0; w=0.3; c=0.1; σ=0.1; budget=0.5	0/0	7.04616	7.870	66.289	65.763	68.489	–	–	–
254	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.3; c=0.1; σ=0.1; POST	0/0	7.09617	23.671	63.699	63.161	64.760	85.548	84.832	89.511
255	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.1; σ=0.1; budget=0.5; POST	0/0	7.03930	17.558	51.099	51.150	45.956	83.713	82.540	44.909
257	A3; G^+ a/m/h	L1N; κ=0; w=0.15	0/0	7.02421	5.072	43.208	41.993	44.173	–	–	–
258	A3; G^+ a/m/h	OL1; κ=0; w=0.15; budget=0.5	0/0	7.01421	5.132	42.818	41.421	46.434	–	–	–
259	A6-POST; G^+ a/m/h/Q/K/V	L1N; κ=0; w=0.15; POST	0/0	7.04048	18.769	46.910	45.985	45.451	89.726	89.749	60.629
260	A6-POST; G^+ a/m/h/Q/K/V	OL1; κ=0; w=0.15; budget=0.5; POST	0/0	7.03898	17.695	50.985	51.021	46.048	88.054	88.050	44.769
261	A3; G^+ a/m/h	L1N; κ=0; w=5	0/0	7.17578	5.182	33.398	28.657	67.441	–	–	–
262	A3; G^+ a/m/h	OL1; κ=0; w=5; budget=0.5	0/0	7.02909	4.708	46.179	27.820	47.595	–	–	–
263	A6-POST; G^+ a/m/h/Q/K/V	L1N; κ=0; w=5; POST	0/0	7.09562	21.319	39.804	26.367	54.307	89.664	89.667	96.086
264	A6-POST; G^+ a/m/h/Q/K/V	OL1; κ=0; w=5; budget=0.5; POST	0/0	7.03870	17.695	51.031	51.066	46.031	88.073	88.098	44.750
266	A3; G^+ a/m/h	RN; κ=0; w=0.1; c=0.1; σ=0.1	0/0	7.03683	7.229	61.069	60.806	62.379	–	–	–
267	A3; G^+ a/m/h	OR; κ=0; w=0.1; c=0.1; σ=0.1; budget=0.5	0/0	7.03121	7.284	61.406	60.934	63.289	–	–	–
268	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.1; c=0.1; σ=0.1; POST	0/0	7.05868	21.882	56.815	56.411	55.271	85.961	85.021	82.347
269	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.1; c=0.1; σ=0.1; budget=0.5; POST	0/0	7.03926	17.547	51.091	51.150	45.981	83.730	82.439	44.862

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Table 13 (continued)

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z_Q^g	z_K^g	z_V^g
270	A3; G^+ a/m/h	RN; κ=0; w=1; c=0.1; σ=0.1	0/0	7.20123	8.772	71.995	72.131	78.937	–	–	–
271	A3; G^+ a/m/h	OR; κ=0; w=1; c=0.1; σ=0.1; budget=0.5	0/0	7.05883	8.251	69.491	69.210	71.555	–	–	–
272	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=1; c=0.1; σ=0.1; POST	0/0	7.16248	24.898	68.029	67.670	73.588	85.179	84.625	92.212
273	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=1; c=0.1; σ=0.1; budget=0.5; POST	0/0	7.03909	17.557	51.090	51.146	46.016	83.821	82.498	44.881
275	A3; G^+ a/m/h	RN; κ=0; w=0.3; c=0.05; σ=0.05	0/0	7.12042	8.361	70.744	70.716	71.682	–	–	–
276	A3; G^+ a/m/h	OR; κ=0; w=0.3; c=0.05; σ=0.05; budget=0.5	0/0	7.03339	7.845	66.217	65.630	68.092	–	–	–
277	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.3; c=0.05; σ=0.05; POST	0/0	7.10916	23.235	63.490	63.171	66.491	81.150	80.633	83.636
278	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.05; σ=0.05; budget=0.5; POST	0/0	7.03514	17.018	51.064	51.134	45.990	74.915	74.655	44.220
279	A3; G^+ a/m/h	RN; κ=0; w=0.3; c=0.5; σ=0.5	0/0	7.07964	7.325	71.014	70.139	47.827	–	–	–
280	A3; G^+ a/m/h	OR; κ=0; w=0.3; c=0.5; σ=0.5; budget=0.5	0/0	7.05550	7.327	70.594	69.427	48.912	–	–	–
281	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.3; c=0.5; σ=0.5; POST	0/0	7.07163	21.009	66.022	64.831	45.239	89.495	89.483	68.828
282	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.5; σ=0.5; budget=0.5; POST	0/0	7.04037	18.100	51.825	51.807	46.323	89.688	89.638	47.707
284	A3; G^+ a/m/h	RN; κ=0; w=0.3; c=0.1; σ=0.05	0/0	7.10143	8.675	74.010	73.782	73.507	–	–	–
285	A3; G^+ a/m/h	OR; κ=0; w=0.3; c=0.1; σ=0.05; budget=0.5	0/0	7.05176	8.243	69.865	69.134	71.156	–	–	–
286	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.3; c=0.1; σ=0.05; POST	0/0	7.09446	24.231	66.765	66.061	66.151	87.652	86.724	91.469
287	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.1; σ=0.05; budget=0.5; POST	0/0	7.03708	17.392	51.022	51.078	45.974	82.656	84.338	44.386
288	A3; G^+ a/m/h	RN; κ=0; w=0.3; c=0.1; σ=0.2	0/0	7.12562	8.087	65.097	65.074	75.157	–	–	–
289	A3; G^+ a/m/h	OR; κ=0; w=0.3; c=0.1; σ=0.2; budget=0.5	0/0	7.04407	7.602	61.910	61.782	69.497	–	–	–
290	A6-POST; G^+ a/m/h/Q/K/V	RN; κ=0; w=0.3; c=0.1; σ=0.2; POST	0/0	7.11203	24.032	61.548	61.391	67.723	88.732	88.746	93.672
291	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.1; σ=0.2; budget=0.5; POST	0/0	7.04009	17.848	51.176	51.231	46.143	88.114	88.059	45.984

A.5 B4: seed-1 sentinel endpoints

Table 14: Canonical S1-B4 core scientific endpoints. R_m and all z columns are percentages; z^g_Q, z^g_K, z^g_V are gate-output zeros.

Cfg	Architecture / gate	Setting	Init/data	Loss	R_m (%)	z_a	z_m	z_h	z^g_Q	z^g_K	z^g_V
293	A0	AdamW; lr=3×10^{-5}	1/1	7.05801	< 0.0001	–	–	–	–	–	–
294	A1-H; G^+ h	AdamW; lr=3×10^{-5}; κ=0	1/1	7.00900	2.086	–	–	48.770	–	–	–
295	A3; G^+ a/m/h	AdamW; lr=3×10^{-5}; κ=0	1/1	7.02508	5.759	50.858	50.911	45.565	–	–	–
296	A6-PRE; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0; PRE	1/1	7.03468	12.319	50.877	51.007	45.842	18.207	27.522	44.464
297	A6-POST; G^+ a/m/h/Q/K/V	AdamW; lr=3×10^{-5}; κ=0; POST	1/1	7.05276	13.852	50.900	50.977	45.832	24.306	35.739	44.311
298	A6-POST; ReLU a/m/h; G^+ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; POST; qkv	1/1	7.05299	17.061	50.713	50.828	46.264	38.405	51.442	66.762
299	A6-POST; ReLU a/m/h; $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; κ=0.1; POST; qkv	1/1	7.02341	11.970	50.624	50.741	45.693	8.862	17.123	48.840
300	A6-POST; ReLU a/m/h; ATG $G^\pm$ Q/K/V	AdamW; lr=3×10^{-5}; learned κ_0=0.1; ABS; layer/site; POST	1/1	7.02248	13.226	50.622	50.762	45.734	13.309	23.333	55.100
301	A6-POST; G^+ a/m/h/Q/K/V	L1N; κ=0; w=1; POST	1/1	7.07032	19.775	42.965	34.761	41.107	89.707	89.697	81.191
302	A6-POST; G^+ a/m/h/Q/K/V	OR; κ=0; w=0.3; c=0.1; σ=0.1; budget=0.5; POST	1/1	7.05829	17.561	50.973	51.026	46.004	83.854	83.485	45.385